SVD = Singular Value Decomposition

Eigenvalue = sum of square of the distances

$$\frac{SS(\text{distances for PC1})}{n - 1} = \text{Eigenvalue for PC1}$$

$$\sqrt{SS(\text{distances for PC1})} = \text{Singular Value for PC1}$$

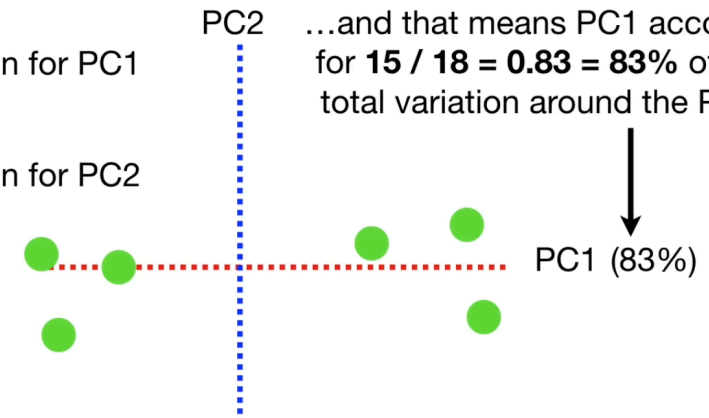
For the sake of the example, imagine that the Variation for **PC1 = 15**, and the variation for **PC2 = 3**.

$$\frac{SS(\text{distances for PC1})}{n - 1} = \text{Variation for PC1}$$

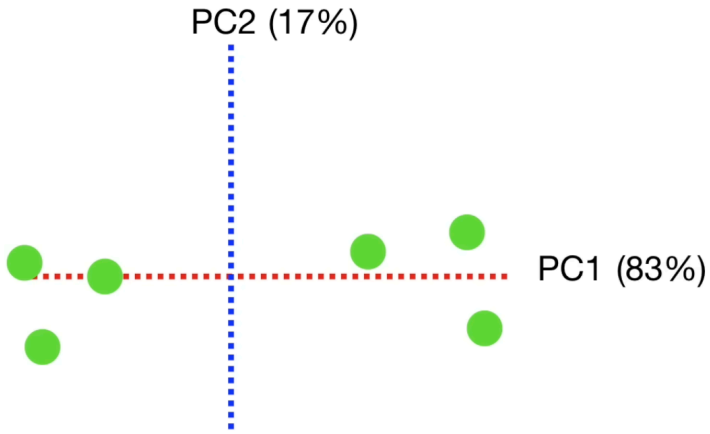
$$\frac{SS(\text{distances for PC2})}{n - 1} = \text{Variation for PC2}$$

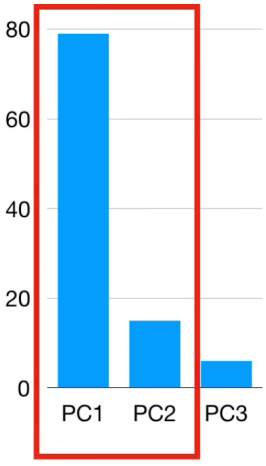
That means that the total variation around both PCs is **15 + 3 = 18**…

…and that means PC1 accounts for **15 / 18 = 0.83 = 83%** of the total variation around the PCs.

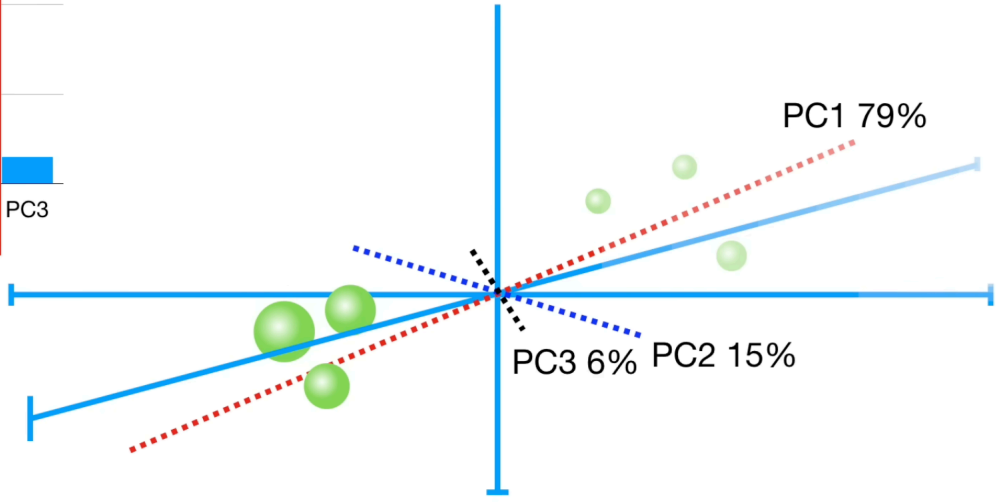


PC2 accounts for **3 / 18 = 0.17 = 17%** of the total variation around the PCs.

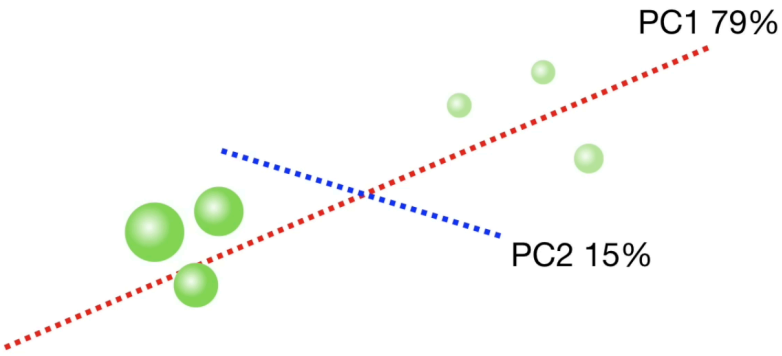




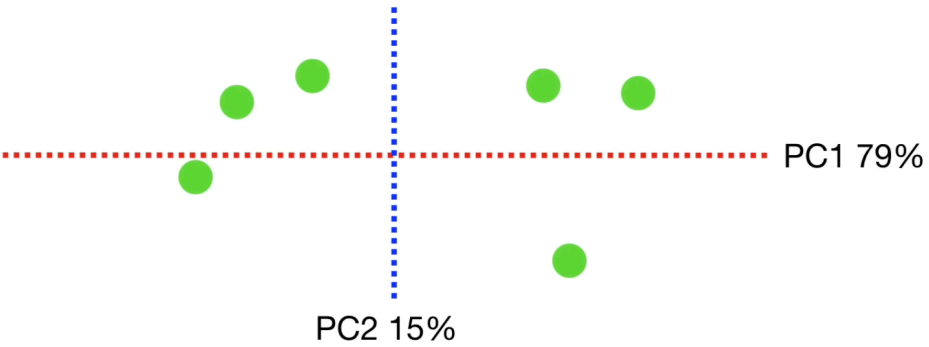
That means that a 2-D graph, using just PC1 and PC2, would be a good approximation of this 3-D graph since it would account for 94% of the variation in the data.

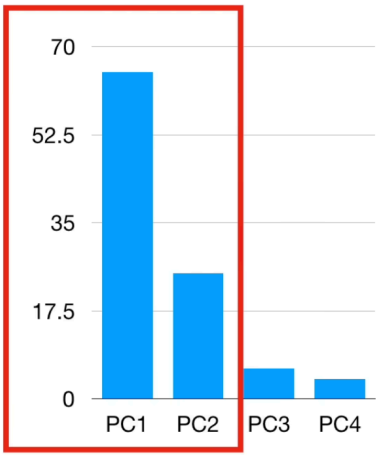


To convert the 3-D graph into a 2-D PCA graph, we just strip away everything but the data and PC1 and PC2…

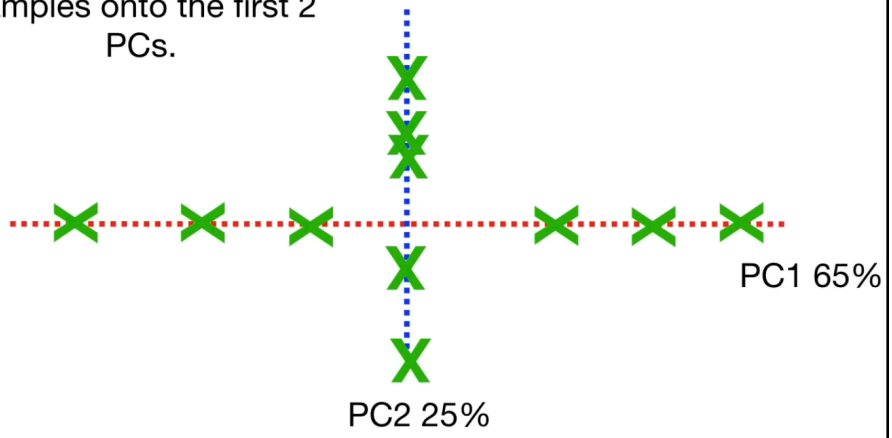


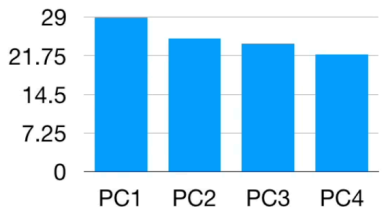
etc…etc…etc…





So we project the samples onto the first 2 PCs.





NOTE: If the scree plot looked like this, where PC3 and PC4 account for a substantial amount of variation, then just using the first 2 PCs would not create a very accurate representation of the data.